

Keng-Yu Chen

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EDUCATION

- PhD Student in Computer Science** Sep. 2024 – Present
* *School of Computer and Communication Sciences, École Polytechnique Fédérale de Lausanne*
* *Thesis Advisor: Serge Vaudenay*
- Master of Science in Computer Science** Sep. 2022 – Jan. 2024
* *Graduate Institute of Electrical Engineering, National Taiwan University*
* *Thesis Advisor: Jiun-Peng Chen & Ho-Lin Chen*
- Bachelor of Science in Computer Science & Information Engineering** Sep. 2018 – Jun. 2022
* *Department of Computer Science & Information Engineering, National Taiwan University*
- (Double Major) Bachelor of Science in Mathematics** Sep. 2018 – Jun. 2022
* *Department of Mathematics, National Taiwan University*

PUBLICATION

1. The Cryptographic Layer of Biometric Authentication
Keng-Yu Chen, Serge Vaudenay
Applied Cryptography and Network Security, 2026
2. When Masking Multiplication Isn't Enough: Exploiting Floating-Point Leakage in Falcon's Pre-Image Computation
Keng-Yu Chen, Ming Qing Ching, Jiun-Peng Chen, Bo-Yin Yang
IACR Transactions on Cryptographic Hardware and Embedded Systems, 2026
3. Masking Floating-Point Number Multiplication and Addition of Falcon
Keng-Yu Chen, Jiun-Peng Chen
IACR Transactions on Cryptographic Hardware and Embedded Systems, 2024

RESEARCH EXPERIENCE

- The Security and Cryptography Laboratory (LASEC)** Sep. 2024 – Present
* *School of Computer and Communication Sciences, École Polytechnique Fédérale de Lausanne*
* *Director: Serge Vaudenay*
- Side-Channel Analysis Laboratory** Jun. 2020 – Feb. 2024
* *Department of Electrical Engineering, National Taiwan University*
* *Director: Jiun-Peng Chen*
- Adjunct Research Assistant** Sep. 2020 – Oct. 2023
* *Research Center for Information Technology Innovation, Academia Sinica*
* *Director: Jiun-Peng Chen*
- Research Intern** Jul. 2023 – Aug. 2023
* *Institute of Information Science, Academia Sinica*
* *Advisor: Kai-Min Chung*

HONORS AND AWARDS

Champion - Innovation Application Competition of Digital Twins for Smart Farming Oct. 2022
* *Council of Agriculture, Executive Yuan, Taiwan*

Academic Achievement Award Apr. 2019
* *Department of Computer Science & Information Engineering, National Taiwan University*

WORKING EXPERIENCE

LASEC Web Admin Sep. 2025 – Present
* *PI: Serge Vaudenay*

Teaching Assistant Jan. 2026 – Jun. 2026
* *COM-506: Student Seminar – Security Protocols and Applications*
* *Teacher: Serge Vaudenay*

EDIC PhD Admission Pre-filtering Dec. 18, 2025

Teaching Assistant Sep. 2025 – Jan. 2026
* *COM-401: Cryptography and Security*
* *Teacher: Serge Vaudenay*

Teaching Assistant Sep. 2020 – Jan. 2021
* *Calculus (1) (2)*
* *Teacher: Ya-Ju Tsai*

SKILLS

Programming Languages: C/C++, Python, Verilog, Makefile, MATLAB, R

Tools: Git/GitHub, ChipWhisperer, L^AT_EX, VIM

Languages: Mandarin (Native), English (IELTS: 7.5)

EXTRACURRICULAR ACTIVITIES

Public Address Team Sep. 2019 – Jun. 2021
* *Student Activity Center, National Taiwan University*
* *Operating stage audio and lighting systems*

Equipment Team Feb. 2019 – Jun. 2021
* *Tainan Alumni Association, National Taiwan University*
* *Operating stage audio and lighting systems, photography, videography*